Reg. No									

St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore – 607001

B.C.A DEGREE EXAMINATION APRIL 2021

CA204S - FUNDAMENTALS OF DATA STRUCTURES

Time : 3 hrs **Max Marks :75**

SECTION - A (5X5=25)

Answer any **FIVE** Questions

1. What is an array data structure? Explain the procedure to insert an element into array.
2. Explain the linear and non-linear data structures.
3. Explain the Maze problem.
4. What is a circular queue?
5. What are the advantages and disadvantages of singly linked list?
6. Give an algorithm for the traversal of circularly linked list.
7. What is a graph? Define directed and undirected graphs with examples.
8. Give an algorithm for depth first search in a Graph.

SECTION - B (5X10=50)

Answer any **FIVE** Questions

9. What are the row-major order and column-major orders? Explain them.
10. Give an algorithm for polynomial addition.

11. Discus the various operations performed on a doubly linked list.
12. What is recursion? Explain with any two applications.
13. Explain the procedure to convert a forest to binary trees and their traversals.
14. Explain the array representation of binary trees with its advantages and disadvantages.
15. Explain the various methods of representation graphs.
16. Give Dijkstra's shortest path algorithm with an example.

****